Diagnostic Expectations and Sovereign Default

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Abstract

Recent news becomes a pricing state when sovereign creditors form diagnostic expectations. In a one-period debt model, equal debt and equal income command different prices after different histories. Good news raises prices and predicts low bond returns under the true probability law. A monthly measurement model maps individual fixed-event forecast revisions into the diagnostic parameter. A five-year bond implementation preserves an exact rational history zero and quantifies the diagnostic effects on spreads, default, and returns. Capital-constrained rational intermediaries attenuate the news slope continuously as their risk-bearing capacity rises. The quantitative exercise imports the published long-debt calibration of Chatterjee and Eyigungor (2012) and reports simulation uncertainty for every principal moment. Debt ceilings raise sovereign welfare when the government extrapolates and lower it when only creditors extrapolate.

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1 Introduction

In June 2017 Argentina sold \$2.75 billion of a hundred-year dollar bond priced to yield 7.9 percent, and investors placed orders for roughly three times that amount. Argentina is a serial defaulter by any standard: Reinhart and Rogoff (2009) count seven external defaults or restructurings since independence, and the 2001 default was then the largest in history. Less than three years after the century bond was sold, in May 2020, the country was in default for the ninth time. A creditor who bought the bond at issue and held it to the restructuring lost most of his investment.

The episode combines a familiar shift in fundamentals with a sharp reversal in creditor demand. Commodity prices, fiscal outcomes, and politics all deteriorated between 2017 and 2020. The bond cycle also belongs to a broader regularity. Emerging-market lending arrives in waves, spreads compress during booms, and optimistic credit conditions forecast low subsequent returns (Greenwood and Hanson, 2013; López-Salido et al., 2017). I study a measurable form of belief movement that connects these return reversals to sovereign default risk.

The workhorse theory of sovereign borrowing, from Eaton and Gersovitz (1981) through its quantitative implementations by Aguiar and Gopinath (2006), Arellano (2008), and Chatterjee and Eyigungor (2012), explains spreads through the option to default in incomplete insurance markets. The theory has real successes. Governments default in bad times, which makes spreads countercyclical. Borrowing remains procyclical. The theory also has a striking property. Every equilibrium object, the price schedule as much as the default set, is a function of the current state, debt b and income y. Two countries with identical debt and identical income borrow on identical terms, whatever histories brought them to that point. Conditional on the state, pricing is memoryless.

The mechanism begins with one departure from the workhorse model, the probability law used by creditors. Creditors hold *diagnostic expectations* in the sense of Bordalo et al. (2018). They overweight future states whose relative likelihood rises with current news. The parameter $\theta \geq 0$ governs the strength of extrapolation, and $\theta = 0$ recovers rational expectations. Under Gaussian income, the diagnostic distribution keeps the true conditional variance and shifts the mean by $\theta\rho(x - h)$, where x is current log income and h is its lag. The quantitative analysis then adds long-duration debt and an explicit survey measurement model without changing this belief law.

The consequence is that one new state variable enters the pricing of sovereign debt: the direction of recent news, $\nu = x - h$. At fixed debt and fixed income, bond prices increase in ν, strictly so wherever default risk is interior (Proposition 2). The rational model predicts a zero coefficient on news once debt and income are controlled for, and the diagnostic model predicts a positive one. That zero is the natural test of the theory. Expected bond returns under the true income process are predictably low after good news and predictably high after bad news (Proposition 3). This is the sovereign analogue of the return predictability that credit-market sentiment generates in corporate bonds. Default regions also inherit history dependence. Debt that is safe after good news becomes default-inducing when the news reverses, so the model produces boom-bust credit cycles from ordinary shocks, outside disasters and sunspot runs (Proposition 4). These comparative statics require monotone default in income, the regularity condition examined below.

The discipline of the exercise is that forecast data pin down θ. Diagnostic expectations imply overreaction in individual forecasts. A forecaster who revises upward tends to be disappointed, so realized forecast errors are negatively related to forecast revisions. Bordalo et al. (2020) document this pattern across United States macroeconomic series in the framework of Coibion and Gorodnichenko (2015). Proposition 5 gives the exact coefficient for a one-quarter fixed target. Consensus forecasts are annual fixed-event objects observed monthly. I therefore simulate that timing and apply the same calendar conversion used on the data. At $\theta = 0.5$, the simulated coefficient is -0.282 rather than the one-quarter value -0.30. Inverting the simulated map returns 0.5. Published estimates place the corresponding degree of overreaction between roughly one-quarter and one (Bordalo et al., 2018; Bordalo et al., 2020). I take $\theta = 0.5$ as the baseline and specify the forecast-revision regressions that estimate it country by country.

The quantitative work separates the short-bond theorem from the long-bond mechanism. The one-period economy uses the Arellano (2008) calibration. At $\theta = 0.5$, one standard deviation of good news lowers the spread by 462 basis points at fixed debt and output, while rational pricing gives zero. The long-bond economy uses the complete published parameter block of Chatterjee and Eyigungor (2012). It adds coupons, dilution, and resale prices. At $\theta = 0$, history remains an exact redundant state. At $\theta > 0$, price monotonicity and the return sign are numerical properties of the computed long-bond equilibrium, reported with fixed-point residuals and simulation uncertainty.

The policy question is then whether governments should bind themselves against belief-driven cycles, and here the model delivers an answer with an edge to it. Welfare depends on *whose* expectations are diagnostic. If the government is rational while the market overreacts, debt limits make the sovereign worse off: the government is selling overpriced claims to optimists and repurchasing them cheap from pessimists, and a ceiling interferes with a trade that benefits it. If the government shares the market's beliefs, the ranking reverses: a ceiling that binds after good-news sequences prevents borrowing the government avoids under the true distribution, and the welfare gain is increasing in θ. I compute the full two-by-two of government and market beliefs. The lesson for the fiscal-rules debate is that rules are a commitment device against the government's own extrapolation rather than a repair of market mispricing.

Niemann and Prein (2025) place diagnostic expectations in a quantitative long-term sovereign debt model, estimate diagnosticity from IMF growth forecasts, and study debt and spread brakes. Those contributions establish the broad quantitative mechanism. I pursue the restrictions that identify diagnostic sovereign pricing. The short-bond theory isolates history dependence at fixed debt and income. The survey block reproduces the monthly timing of individual fixed-event forecasts. The central market test uses future sovereign bond returns. The intermediary extension shows how rational risk-bearing capacity attenuates the news slope. The two belief assignments in the policy exercise separate gains from creditor mispricing from losses caused by government extrapolation. These restrictions distinguish the paper from an additional long-debt calibration of diagnostic expectations.

The remaining comparison is with rational sovereign default models and the wider evidence on belief distortions. Quantitative default models explain countercyclical spreads through default incentives (Aguiar and Gopinath, 2006; Arellano, 2008; Chatterjee and Eyigungor, 2012; Mendoza and Yue, 2012). Multiplicity models generate history dependence through equilibrium selection (Cole and Kehoe, 2000; Lorenzoni and Werning, 2019). Diagnostic beliefs instead give public news a signed return implication. Forecast research measures the same extrapolation outside asset prices (Bordalo et al., 2020). Combining the forecast measure with future bond returns supplies the empirical discipline.

My plan is as follows. Section 2 presents a two-period example that contains the paper's main mechanisms in closed form. Section 3 sets out the full model and defines equilibrium. Section 4 states the theoretical results. Section 5 maps survey overreaction to θ and specifies the measurement design. Section 6 reports the quantitative results, Section 7 the spread-panel evidence and its design, and Section 8 the fiscal-rule experiments. Section 9 concludes. Proofs and extensions are in the appendix.

2 A Two-Period Example

Nearly everything in this paper can be seen in a two-period economy, so I begin there. A sovereign has endowment y_0 at date 0 and a random endowment at date 1 that takes the value y_H with true probability π and $y_L < y_H$ otherwise. At date 0 it sells one-period bonds with face value b to competitive risk-neutral foreign lenders whose opportunity cost is the gross rate R. At date 1 it repays or defaults. Default costs a fraction ψ of that period's endowment, so the sovereign repays exactly when $b \leq \psi y_1$. I restrict attention to the interesting region $\psi y_L < b \leq \psi y_H$, where debt is repaid in the high state and repudiated in the low state.

Lenders use a distorted probability. At date 0 they have just observed news about the country: formally, the current probability π differs from a reference probability π_0 that summarizes beliefs before the news arrived. Diagnostic lenders overweight the states the news has made more representative. I take the linear form

$$\hat{\pi} \;=\; \pi + \theta\,(\pi - \pi_0), \qquad \theta \geq 0, \tag{1}$$

which is the two-state counterpart of the density tilt introduced in Section 3. The news is $\nu = \pi - \pi_0$, and $\theta = 0$ recovers rational expectations. Competitive risk-neutral pricing gives the bond price

$$q \;=\; \frac{\hat{\pi}}{R} \;=\; \frac{\pi + \theta\nu}{R}. \tag{2}$$

Equation (2) already contains the paper's first result. Fix fundamentals: two countries with the same current repayment probability π, the same debt, the same output. If one arrived at π from below (good news, $\nu > 0$) and the other from above, the first borrows more cheaply. History matters for prices even though it is payoff-irrelevant for repayment, because history determines what the news was, and news is what diagnostic beliefs overweight.

The second result concerns returns. A lender who buys at price q receives 1 with true probability π, so his expected gross return is $\pi/q = R\,\pi/\hat{\pi}$, and the expected excess return is

$$\mathbb{E}[R^e] \;=\; R\,\frac{\pi - \hat{\pi}}{\hat{\pi}} \;=\; -\,\frac{R\,\theta\nu}{\pi + \theta\nu}. \tag{3}$$

After good news, creditors systematically overpay and earn low returns. After bad news the reverse holds. At $\theta = 0$ the excess return is zero for every ν: predictable returns from public news are exactly the rational model's forbidden pattern, which is what makes them evidence.

The third result concerns quantities. Let the sovereign have log utility, discount factor β, and rational beliefs π. Only the market is diagnostic for now. It chooses b to maximize $\ln(y_0 + qb) + \beta[\pi \ln(y_H - b) + (1-\pi)\ln((1-\psi)y_L)]$. The first-order condition gives, after rearrangement,

$$b^* \;=\; \frac{y_H}{1+\beta\pi} \;-\; \frac{\beta\pi R\,y_0}{\hat{\pi}\,(1+\beta\pi)}, \tag{4}$$

which is strictly increasing in $\hat{\pi}$ and hence in the news ν. Optimism the government treats as market pricing still moves the sovereign: a higher $\hat{\pi}$ raises the price at which it can sell claims on the high state, and it rationally sells more of them. Borrowing responds to market sentiment about fundamentals over and above fundamentals themselves. There is also a margin in ν at which the sovereign switches from safe debt ($b \leq \psi y_L$) to risky debt, since the value of the risky plan rises with $\hat{\pi}$ while the safe plan's value is independent of it. Optimism moves the economy from a region where default never happens to one where it happens with probability $1 - \pi$. Good news, filtered through diagnostic pricing, buys fragility.

Whose welfare does the distortion damage? Here the example teaches a lesson that survives, quantitatively, in the full model. The rational sovereign in (4) gains from mispricing: it sells overpriced claims after good news, and the expected losses in (3) fall on the lenders. Capping b would deny the sovereign a profitable trade. The case for a borrowing limit appears only when the government itself prices the future with $\hat{\pi}$ rather than π, so that its own plans embody the extrapolation. Then a rule chosen behind the veil, before the news arrives, can raise the government's welfare evaluated under the truth. Section 8 makes this precise and puts numbers on it.

The example isolates one margin. The full model adds history-dependent default sets, booms that build over several quarters, and the equilibrium recursion that maps future default behavior into today's prices.

3 The Model

3.1 Environment

Time is discrete and infinite. A small open economy receives a stochastic endowment y_t whose logarithm $x_t = \log y_t$ follows

$$x_{t+1} = \rho x_t + \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim N(0, \sigma_\varepsilon^2), \qquad 0 < \rho < 1. \tag{5}$$

The government is benevolent, with preferences $\mathbb{E}_0 \sum_t \beta^t u(c_t)$ and $u(c) = c^{1-\gamma}/(1-\gamma)$. It borrows by selling one-period non-contingent bonds to competitive, risk-neutral foreign lenders whose outside option is the risk-free gross rate $R = 1 + r^*$. A bond with face value b' sold today promises b' tomorrow. The government chooses repayment sequentially, so equilibrium default prices sustain sovereign debt.

At the start of a period the government owes $b \geq 0$ and chooses whether to repay or default. Under repayment it pays b and sells new debt $b' \in \mathcal{B} \subset \mathbb{R}_+$ at the market price q, giving consumption $c = y + qb' - b$. Default writes off the debt, excludes the country from markets, and imposes the output loss standard in this literature: income during exclusion is $y^D(x) = \min\{y, \phi\,\mathbb{E}[y]\}$. This asymmetric cost bites in good states and gives realistic default frequencies in Arellano (2008). Mendoza and Yue (2012) derive it from imported-input substitution. Each period, exclusion ends with probability λ and the country re-enters with zero debt.

All of this is the workhorse model of Eaton and Gersovitz (1981) and Arellano (2008). The single departure is next.

3.2 Creditor beliefs

Lenders form beliefs about x' by distorting the true conditional density $p(x' \mid x)$ toward states that recent news has made more representative. Let $h = x_{-1}$ denote last period's log income. Following Bordalo et al. (2018), the diagnostic density is

$$\hat{p}_\theta(x' \mid x, h) \;=\; p(x' \mid x) \left[\frac{p(x' \mid x)}{p(x' \mid h)}\right]^\theta \frac{1}{Z(x,h)}, \qquad \theta \geq 0, \tag{6}$$

where Z is the normalizing constant. The reference density $p(\cdot \mid h)$ is the forecast an analyst would have made had the economy stood still: the distortion overweights precisely those states whose odds have improved relative to that reference, which is the "kernel of truth" property. At $\theta = 0$ beliefs are rational. The distortion is a news-contingent amplifier rather than pessimism or optimism as a trait.

For the Gaussian autoregression (5) the tilt takes a form that makes the model tractable.

Lemma 1 (Diagnostic tilt)**.** *Let $p(\cdot \mid x)$ be the $N(\rho x, \sigma_\varepsilon^2)$ density and $p(\cdot \mid h)$ the $N(\rho h, \sigma_\varepsilon^2)$ density. Then $\hat{p}_\theta(\cdot \mid x, h)$ is the $N\big(\rho x + \theta\rho(x - h),\, \sigma_\varepsilon^2\big)$ density.*

The proof, a matter of completing a square, is in Appendix A. Diagnostic creditors carry the same conditional variance and the wrong mean. The mean distortion is proportional to the news, $\nu \equiv x - h$: after a quarter in which log income rose, creditors expect the rise to continue beyond what persistence justifies. The belief state is only the pair (x, h), so the departure from rational expectations costs one state variable that is observable as a growth realization or a forecast revision.

Diagnostic lenders price the marginal unit of emerging-market debt. Individual professional analysts, the population represented in the model, revise too far in the direction of news (Bordalo et al., 2020; Bordalo et al., 2022). An arbitrageur who trades against this distortion must carry a large position and bear default risk until beliefs correct. Limited intermediary capital therefore lets diagnostic beliefs enter competitive prices (Greenwood and Hanson, 2013; López-Salido et al., 2017). Risk-neutral diagnostic pricing supplies the behavioral counterpart of rational risk-neutral pricing in the workhorse model. The baseline keeps the government rational and assigns overreaction to the market, which isolates the pricing distortion. Section 8 then completes the two-by-two of government and market beliefs for normative analysis.

3.3 Recursive formulation and equilibrium

The aggregate state for pricing is (b', x, h). The government's value satisfies

$$V(b, x, h) = \max\Big\{V^R(b, x, h),\, V^D(x)\Big\}, \tag{7}$$

with repayment value

$$V^R(b, x, h) = \max_{b' \in \mathcal{B}} \Big\{u\big(y + q(b', x, h)\, b' - b\big) + \beta\, \mathbb{E}\big[V(b', x', x) \,\big|\, x\big]\Big\}, \tag{8}$$

where the expectation is under the true kernel (5) because the government is rational, and default value

$$V^D(x) = u(y^D(x)) + \beta\,\mathbb{E}[\lambda V(0, x', x) + (1-\lambda)V^D(x') \,|\, x]. \tag{9}$$

Note the bookkeeping in the continuation values: tomorrow's reference point is today's income, $h' = x$. Let $d(b, x, h) \in \{0, 1\}$ indicate default. Lenders break even under *their* beliefs:

$$q(b', x, h) \;=\; \frac{1}{R}\,\widehat{\mathbb{E}}_\theta\big[\,1 - d(b', x', x) \,|\, x, h\big], \tag{10}$$

where $\widehat{\mathbb{E}}_\theta$ integrates against $\hat{p}_\theta(\cdot \mid x, h)$. Equation (10) is where history enters: h appears in the price because h shapes the belief, even though h is payoff-irrelevant for x' under the truth.

Definition 1. A recursive equilibrium with diagnostic pricing is a value function V, policies d and $b'(\cdot)$, and a price schedule q such that (i) given q, the value and policies solve (7)–(9), and (ii) given d, the price satisfies (10) for all (b', x, h).

This is a temporary-equilibrium concept: agents optimize under their stated beliefs and markets clear. The true measure evaluates outcomes. Discrete debt choices make deterministic policies jump at ties. Mixed actions convexify those ties and deliver finite-economy existence.

Proposition 1 (Finite-grid existence)**.** *Let the income and debt grids be finite and let $0 < \beta < 1$. Extend utility below a positive consumption floor by its tangent at the floor, so current returns remain finite and continuous for every grid action and price. Then a stationary recursive equilibrium exists in mixed government policies. A computed deterministic fixed point whose repayment choices lie above the floor is an equilibrium of the original positive-consumption economy whenever its Bellman and zero-profit residuals are zero.*

The proof in Appendix A applies Kakutani's theorem to prices and mixed stationary policies. The reported deterministic equilibrium satisfies Bellman and price residuals below 5×10^{-8}. Iteration from risk-free prices, zero prices, and the rational price schedule produces price arrays whose largest difference is 10^{-6}. At $\theta = 0$, h drops out of (10) and the model is exactly Arellano (2008).

The question the theory must answer is what the extra state variable does to prices, returns, and default sets. That is the next section.

4 Theoretical Results

Throughout, $\nu = x - h$ denotes the news and $F_\theta(\cdot \mid x, h)$ the diagnostic conditional cdf. The structure of the results is as follows. Lemmas 1 and 2 and Proposition 5 are exact properties of the belief kernel alone. Lemma 3 proves the debt margin of monotone default in any equilibrium. Propositions 2 to 4 are comparative statics under Assumption 1, the income margin of monotone default, verified at every computed equilibrium across grid densities, default-cost specifications, and parameter vectors (Appendix C). The borrowing response to news and the welfare magnitudes of Section 8 are numerical properties of the computed equilibria.

Lemma 2 (News orders beliefs)**.** *Fix x. If $\nu_2 > \nu_1$ then $F_\theta(\cdot \mid x, h_2)$ first-order stochastically dominates $F_\theta(\cdot \mid x, h_1)$, where $h_i = x - \nu_i$, with strict dominance whenever $\theta > 0$.*

Lemma 3 (Default is monotone in debt)**.** *In any equilibrium of Definition 1, for every (x, h): the repayment value $V^R(b, x, h)$ is nonincreasing in b, strictly decreasing wherever finite, $V^D(x)$ is independent of b, and the default set $\{b \in \mathcal{B} : d(b, x, h) = 1\}$ is an upper interval $\{b \geq \underline{b}(x, h)\}$.*

Lemma 3 is unconditional: a larger inherited debt lowers consumption at every issuance choice and leaves the continuation problem unchanged, while the default value is independent of b. The short proof is in Appendix A. The income margin is the paper's single maintained regularity condition.

Assumption 1 (Monotone default in income)**.** *The default policy $d(b, x, h)$ is nonincreasing in x for fixed (b, h).*

Assumption 1 says default is a bad-times event. Arellano (2008) proves it for i.i.d. income. With persistent income it is the maintained regularity condition of the quantitative literature even at $\theta = 0$, because repayment and default values both rise with income and their ranking is state dependent. The verification is systematic: the condition holds at every grid point of every computed equilibrium at $\theta \in \{0, 0.5\}$, it continues to hold when both grids are doubled ($n_x = 31$, $n_b = 400$), under a proportional default cost in place of the asymmetric one, and at every parameter vector visited by the recalibration search of Section 6. Appendix C reports the counts, and the small 0.1 percent of columns at $\theta = 1$ are traceable to near-degenerate pricing at extreme states.

The next lemma gives a restricted sufficient condition and locates the economic force behind the assumption.

Lemma 4 (A sufficient condition for income-monotone default)**.** *Consider the one-period economy with i.i.d. income and proportional default consumption $y^D = (1 - \psi)y$, where $0 < \psi < 1$. Fix inherited debt and any belief state. Suppose every feasible repayment plan satisfies $c^R(y, b') \leq \bar{c}(y)$ and*

$$u'(\bar{c}(y)) \geq (1 - \psi)u'((1 - \psi)y)\,.$$

Then $V^R(b, y) - V^D(y)$ is nondecreasing in y, so default is nonincreasing in income. Under CRRA utility with coefficient $\gamma > 0$, the displayed condition is equivalent to

$$\bar{c}(y) \leq (1 - \psi)^{(\gamma - 1)/\gamma} y.$$

With linear utility it holds for every feasible repayment plan.

The condition requires the marginal value of current income under repayment to exceed its marginal value in default. It holds automatically with linear utility and holds under CRRA whenever debt service keeps repayment consumption below the stated bound. The persistent-income quantitative model lies outside this restricted case, so the grid audit continues to discipline the broader specification.

Proposition 2 (History-dependent prices)**.** *Under Assumption 1, the price schedule* $q(b', x, h)$ *is nondecreasing in* ν *for every* (b', x)*. It is strictly increasing between any two news values when* $\theta > 0$ *and the repayment indicator differs on a set with positive diagnostic probability. Equivalently, strictness requires*

$$0 < \widehat{\mathbb{P}}_\theta\{d(b', x', x) = 0 \mid x, h\} < 1.$$

Corollary 1 (The rational zero)**.** *If* $\theta = 0$*, prices and spreads are invariant to* ν *conditional on* (b', x)*. If* $\theta > 0$*, the price is strictly increasing in* ν *in every cell satisfying the interior-risk condition of Proposition 2. Thus the rational benchmark predicts a zero within-cell news coefficient, while the diagnostic model predicts a negative spread coefficient in interior-risk cells.*

Proposition 2 is the paper's pricing result and Corollary 1 its identification content. Debt and output are controls and the news is the treatment. The workhorse model supplies the sharp null. Spreads still move with growth under both hypotheses through x itself. The diagnostic test lives entirely inside the conditional.

Proposition 3 (Predictable returns)**.** *Let* $R^e(b', x, h)$ *denote the true-measure expected gross return on a positive-price bond in excess of* R*. Under Assumption 1,* R^e *is nonincreasing in* ν*. It equals zero for every* ν *when* $\theta = 0$*. When* $\theta > 0$*, it is nonpositive for* $\nu > 0$ *and nonnegative for* $\nu < 0$*, with strict inequalities under the interior-risk condition of Proposition 2.*

The mechanism is the one in equation (3): after good news the lenders' repayment probability exceeds the true one, so they overpay. Proposition 3 turns the sentiment-reversal evidence from corporate credit (Greenwood and Hanson, 2013; López-Salido et al., 2017) into a prediction for sovereign bonds, with the news direction as the conditioning variable.

Proposition 4 (Bust fragility)**.** *Under Assumption 1, the repayment value* $V^R(b, x, h)$ *is nondecreasing in* ν *and* $V^D(x)$ *is independent of* ν*. Hence the default set* $D(x, h) = \{b : d(b, x, h) = 1\}$ *is weakly decreasing in* ν*: any debt level that triggers default after good news triggers it after bad news, and the implication is one-way.*

Proposition 4 is where the model earns the word *cycle*. A sovereign that accumulated debt while $\nu > 0$, at prices that made the debt look cheap, faces a weakly larger default region when ν turns negative at the same debt and income. The inclusion is strict only when the price change moves repayment value across default value for some debt. The mechanism does not require a sunspot, in contrast to Cole and Kehoe (2000) and Lorenzoni and Werning (2019). The boom can create its own fragility.

Borrowing behavior along the boom completes the account. Whether issuance rises with ν involves an income effect: better prices raise current consumption, which lowers the marginal utility that borrowing serves. In the two-period example the closed form (4) settles the question, and in the computed equilibria of Section 6 issuance is increasing in ν throughout the relevant region. The result is a quantitative property of the computed economy.

These results say what diagnostic pricing does. Whether it does so *in fact* depends on a single number, θ. The next section measures θ outside the spread data entirely.

5 Measuring Overreaction

5.1 Forecast errors and revisions

Diagnostic expectations leave fingerprints in forecast data. Consider a forecaster predicting x_{t+1} under the beliefs of Section 3.2, and define the forecast revision $\mathrm{Rev}_t = \hat{\mathbb{E}}_t[x_{t+1}] - \hat{\mathbb{E}}_{t-1}[x_{t+1}]$ and the forecast error $\mathrm{FE}_{t+1} = x_{t+1} - \hat{\mathbb{E}}_t[x_{t+1}]$. Coibion and Gorodnichenko (2015) proposed regressing errors on revisions:

$$\mathrm{FE}_{t+1} = \alpha + \beta_{CG}\,\mathrm{Rev}_t + u_{t+1}. \tag{11}$$

Under rational expectations $\beta_{CG} = 0$: revisions are part of the information set. Under information rigidity, consensus forecasts underreact and $\beta_{CG} > 0$, which is what Coibion and Gorodnichenko (2015) find for consensus data. Under diagnostic expectations, individual forecasts overreact: an upward revision predicts disappointment, $\beta_{CG} < 0$, which is what Bordalo et al. (2020) find for individual professional forecasters across most United States macroeconomic series. The two findings coexist because averaging across forecasters with dispersed information pushes the consensus coefficient up. The individual-level coefficient is the one that identifies θ, and individual-level data are therefore essential: a consensus-based estimate would confound overreaction with the mechanical underreaction of averages and could even reverse its sign.

For a one-period fixed target at the model frequency, the mapping is analytic.

Proposition 5 (Measurement mapping)**.** *Let x_t follow (5) and let one-period-ahead forecasts be diagnostic with parameter θ, with revisions defined on the fixed target x_{t+1}. Then the population coefficient in (11) is*

$$\beta_{CG} = -\frac{\theta\,(1+\theta)}{(1+\theta)^2 + \theta^2\rho^2}. \tag{12}$$

The derivation is in Appendix A. The function is zero at $\theta = 0$ and strictly decreasing on the parameter region used here. It is the correct inversion only for a one-period fixed target sampled at the model frequency. At $\rho = 0.945$, $\theta = 0.25$ gives $\beta_{CG} = -0.19$, $\theta = 0.5$ gives -0.30, and $\theta = 1$ gives -0.41. These values provide an analytic benchmark. They are not applied mechanically to annual fixed-event forecasts.

5.2 Design for emerging-market forecasts

The estimates just cited are for the United States. The model is about emerging markets, and the discipline requires θ estimated from forecasts of the same economies whose bonds are being priced. I use $\theta = 0.5$ as the baseline value and build the measurement so that the same panel delivers country-level θ_i for the spread tests below.

Consensus Economics surveys professional forecasters on current-year and next-year real GDP growth. The survey records these fixed-event forecasts each month. Let s_t be the number of months remaining in the current year. The fixed-horizon object is

$$F_{ijt}^{12} = \frac{s_t}{12} F_{ijt}^{CY} + \left(1 - \frac{s_t}{12}\right) F_{ijt}^{NY}.$$

The empirical revision is the month-to-month change in this object. The realized counterpart uses first-release annual growth and the same calendar weights. The estimating equation at the forecaster, country, and month level is

$$\mathrm{FE}_{i,j,t} \; = \; \alpha_i + \tau_t + \beta_{CG}\,\mathrm{Rev}_{i,j,t} + u_{i,j,t}, \tag{13}$$

with country and month effects. The coefficient is mapped to θ by simulation, not by (12). I simulate the monthly income process for 240000 months, construct current-year and next-year forecasts at every survey date, apply the calendar conversion above, and estimate (13). At the baseline input $\theta = 0.5$, the resulting population analogue is -0.282. Numerical inversion returns 0.5, with error 0.0. The map is strictly decreasing over $\theta \in [0, 1]$ in the simulation. Appendix C reports the algorithm and recovery grid.

Forecast reporting error enters the forecast error and revision with opposite signs and creates a negative mechanical covariance. The instrument uses predetermined heterogeneity in how forecasters respond to scheduled public releases. For each forecaster, I estimate the historical loading on a release category using observations dated before the target forecast. The interaction of that loading with the current release surprise instruments the individual revision. The regression includes the release surprise itself, country-by-release effects, and the forecast level, so common news enters directly while identification comes from differential pre-period responsiveness. An errors-in-variables correction based on repeated reports supplies a second estimate.

Country-level diagnosticity enters the pricing tests through cross-fitting. One fold estimates the monthly revision map and θ_i. A disjoint fold estimates spread and return responses. Repeating the split generates the joint sampling distribution of both stages. A hierarchical specification shrinks noisy country coefficients toward the regional distribution and integrates over their uncertainty when estimating the cross-country restriction. Moving-block resampling by country preserves overlapping forecast targets and serially correlated bond returns.

6 Quantitative Analysis

6.1 One-period benchmark and mechanism experiment

The one-period model isolates the theorem's mechanism. I solve it on the quarterly calibration of Arellano (2008) for Argentina, reproduced in Table 1. I hold every parameter outside beliefs fixed and vary $\theta \in \{0, 0.5, 1\}$. The resulting columns are controlled comparative statics. The income process is a 21-state Tauchen (1986) chain and debt has 200 points. The value function and price schedule are iterated jointly to a tolerance of 5×10^{-8}. The diagnostic Gaussian is integrated on the same income grid. The fixed-point audit starts prices at the risk-free schedule,

Table 1: Calibration (quarterly).

Parameter		Value	Category	Source or target
Risk aversion	γ	2	externally fixed	standard value in this literature
Discount factor	β	0.953	internally calibrated	default moments at $\theta = 0$ (A08)
Risk-free rate	r^*	0.017	externally fixed	U.S. five-year bond yield (A08)
Income persistence	ρ	0.945	directly estimated	detrended Argentine real GDP (A08)
Income innovation s.d.	σ_ε	0.025	directly estimated	detrended Argentine real GDP (A08)
Re-entry probability	λ	0.282	externally fixed	market re-access delays (A08)
Default output cost	ϕ	0.969	internally calibrated	spread moments at $\theta = 0$ (A08)
Overreaction	θ	$\{0, 0.5, 1\}$	published range	Section 5

Notes: A08 is Arellano (2008). The category column records whether a parameter is externally fixed, directly estimated, internally calibrated, or set by the belief-measurement exercise. Parameters marked internally calibrated at $\theta = 0$ are re-selected at $\theta = 0.5$ in Section 6.6. Table 7 in Appendix C reports units, equation locations, sensitivity ranges, and headline-result effects for every row.

Table 2: Long-duration calibration, quarterly

Parameter	Value	Status	Source or target
Risk aversion γ	2	externally fixed	sovereign-default convention
Discount factor β	0.95460	jointly calibrated	debt and spread moments
Default cost d_0	−0.18845	jointly calibrated	debt and spread moments
Default cost d_1	0.24559	jointly calibrated	debt and spread moments
Income persistence ρ	0.948503	estimated	Argentine real GDP
Innovation s.d. σ_ε	0.027092	estimated	Argentine real GDP
Temporary shock s.d. σ_m	0.003	Chatterjee and Eyigungor (2012)	common across belief specifications
Maturity rate δ	0.05	externally fixed	five-year median maturity
Quarterly coupon z	0.03	externally fixed	12 percent annual coupon
Re-entry probability ξ	0.0385	externally fixed	Argentine exclusion spells
Risk-free rate r^*	0.01	externally fixed	real three-month Treasury rate
Overreaction θ	0.5	measured	monthly forecast revisions

the zero schedule, and the rational equilibrium. The three solutions agree to less than 10^{-6}, and the largest Bellman or pricing residual is 5×10^{-8}.

6.2 Long-duration quantitative model

The main quantitative implementation replaces the one-quarter bond by the five-year random-maturity bond in Appendix E.2. Table 2 reports the parameter block. Chatterjee and Eyigungor (2012) match the maturity rate and coupon to Argentine bonds, estimate the income process from Argentine output, and set re-entry from observed exclusion spells. They select (β, d_0, d_1) jointly to match mean debt, the mean spread, and spread volatility. Moving this block together preserves its economic meaning.

The reported long-debt exercise transports the published parameter block and changes only θ. This external calibration keeps maturity, default technology, patience, and income risk tied to one internally coherent source. It also gives the comparison a direct interpretation. Every difference between the two columns of Table 3 comes from the probability law used by lenders.

The replication code also defines a six-moment SMM estimator for a matched country panel.

Table 3: Long-duration mechanism experiment

	$\theta = 0$		$\theta = 0.5$	
	Estimate	S.E.	Estimate	S.E.
Median annual spread, percent	5.33	0.02	6.48	0.02
Spread s.d., percent	3.57	0.02	4.04	0.02
Defaults per 100 years	3.59	0.04	1.91	0.03
Mean debt to output, percent	70.8	0.09	68.4	0.06
Debt service to output, percent	5.6	0.01	5.4	0.0
Within-cell news coefficient, bp per s.d.	0.0	0.0	-126	0.8
Expected excess return after good news, bp	0.0	0.0	-243.0	0.9
Expected excess return after bad news, bp	0.0	0.0	722.6	4.3

Notes: the table holds the published long-debt parameter block fixed across columns. Standard errors use 250 nonoverlapping batches of 2,000 simulated quarters and preserve the model's serial dependence. The rational return entries equal zero up to the reported fixed-point tolerance.

It takes the target covariance matrix as an input, reports scale-diagonal and covariance-optimal weights, computes the moment Jacobian by branch-preserving continuation, tests its local rank, and forms sandwich intervals for (β, d_0, d_1). The available published moments support the transported calibration. A matched country panel and its joint moment covariance activate the SMM stage. The results below use the published parameter block and attach uncertainty to every simulated outcome.

Long debt changes the return object. A lender receives principal, coupon, and the resale price of the surviving claim. The rational history zero remains exact. At $\theta = 0.5$, the computed equilibrium produces a negative expected excess return after good news and a positive return after bad news. These long-bond signs are numerical properties rather than corollaries of Proposition 3, since resale prices and future issuance enter the payoff. The one-period result isolates the theorem without dilution.

6.3 Prices and default sets

Figure 1 shows the equilibrium price schedule $q(b', x, h)$ at median income for three histories: good news, neutral news, and bad news, each against the rational schedule, which is a single curve because it is independent of h. The histories differ only in where income stood last quarter. Current fundamentals are identical. Diagnostic pricing fans the schedule out. After good news the sovereign can carry more debt at any price, and the maximum revenue it can raise from the market, $\max_{b'} qb'$, is larger. After bad news the schedule collapses inward, and debt levels that were financeable a quarter ago cease to be financeable. The vertical distance between the good-news and bad-news curves at intermediate debt is the price of history, and it is large precisely in the region where default risk is interior, as Proposition 2 says it must be.

Figure 2 translates prices into default behavior: it plots the smallest debt at which the sovereign defaults, as a function of income, for the same three histories. This is Proposition 4

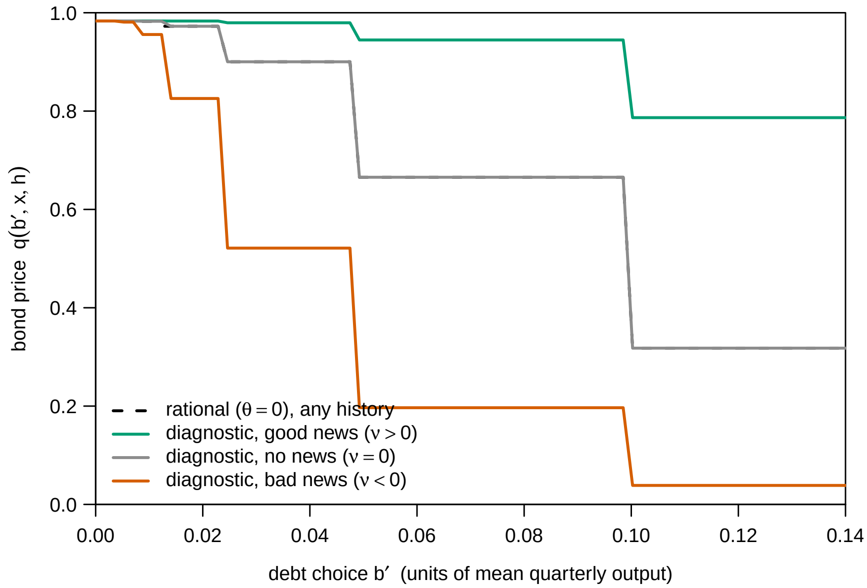

Figure 1: Bond price schedules at median income, $\theta = 0.5$, for three histories of news $\nu = x - h$ (good and bad news set $|\nu|$ to about 2.7 standard deviations of the quarterly innovation for visibility). The dashed curve is the rational schedule, common to all histories.

drawn. Between the good-news and bad-news cutoffs lies the fragility corridor: combinations of debt and income that are safe if the economy arrived on an upswing and default-inducing if it arrived on a downswing. A sovereign whose debt sits in the corridor is betting on the sign of the next quarter's news. The rational model collapses this corridor to the empty set.

6.4 Simulated moments

Table 4 reports moments from 500,000 simulated quarters under the *true* income process, the same shock sequence for every θ.

The news coefficient is zero to machine precision at $\theta = 0$. This is Corollary 1 holding in the simulated data, and the zero requires nonparametric conditioning. A linear regression of spreads on news with linear debt and output controls yields -168 basis points per standard deviation in the *rational* economy, an artifact of the spread schedule's curvature loading on the news term. At $\theta = 0.5$ the same linear specification even gets the sign wrong, because good news shifts issuance along the schedule. Within cells, the diagnostic economy prices news at -462 basis points per standard deviation and the rational economy at zero. The empirical test in Section 7 therefore conditions flexibly on debt and output.

The workhorse model's known weakness with one-period debt is that spreads are too stable. Diagnostic pricing multiplies the standard deviation of spreads by roughly 8.6 while *lowering* the median spread from 3.6 to 1.6 percent. The economy spends most of its time calm, with compressed spreads, and occasionally explodes. Long quiet spells punctuated by crises emerge here from Gaussian shocks alone, outside disasters and regime switches.

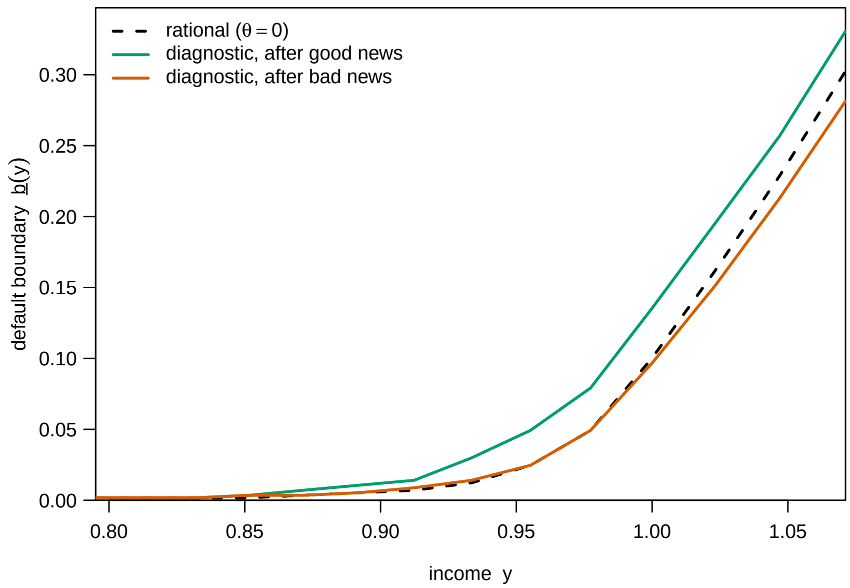

Figure 2: Default cutoffs $\underline{b}(y)$: the smallest debt triggering default, by income, $\theta = 0.5$. The region between the good-news and bad-news curves is history dependent.

The default frequency rises from 3.7 to 11.6 per century, against 2–3 per century in Argentine data. Arellano (2008) selected the parameters outside beliefs to match default frequency at $\theta = 0$. Table 4 therefore traces the fixed-parameter comparative statics in θ. Section 6.6 re-selects (β, ϕ) at the baseline θ and reports the surviving moments. Propositions 2–4 continue to govern every parameter vector satisfying Assumption 1. At $\theta = 1$, the economy loses market access a quarter of the time. This stress calculation establishes the value of estimating θ inside each sovereign market.

6.5 Booms that end

The model's dynamic content is easiest to see in an event study, and the simulation design allows a clean one: because the income path is exogenous and the shock sequence is common across θ, I can locate every episode, 5,593 of them in half a million quarters, in which log income rises for at least three consecutive quarters and then falls, and compare the rational and diagnostic economies *on the same episodes*. Figure 3 shows median spreads, mean debt, and the default hazard in event time.

The figure shows the whole belief cycle. Entering the episode, on the heels of whatever bad news preceded the boom, the diagnostic economy pays spreads several times the rational level: pessimism is extrapolated too. The boom then compresses diagnostic spreads *below* the rational benchmark, quarter by quarter, while debt rises by 136 percent against 123 percent in the rational economy: the diagnostic sovereign borrows into the optimism, as the two-period example said it rationally should. The difference arrives with the reversal. In the quarter the news turns, the default hazard reaches 12.6 percent in the diagnostic economy, while on the

Table 4: Simulated moments under the true income process.

	$\theta = 0$	$\theta = 0.5$	$\theta = 1$
Median spread (percent, annualized)	3.6	1.6	0.5
S.d. of spread (percent)	6.8	57.8	1,277.6
Correlation of spread with log income	-0.29	-0.17	-0.28
Mean debt service to output (percent)	4.4	3.8	6.1
Defaults per 100 years	3.7	11.6	29.3
Share of quarters with market access (percent)	96.7	89.7	73.9
News coefficient, within (b', y) cells (bp per 1σ)	0.0	-462	$-21{,}313$
Implied forecast-revision coefficient β_{CG}	0	-0.30	-0.41

Notes: spreads are annualized from one-period bond prices and winsorized at the 1st and 99th percentiles before computing means and standard deviations. Medians are unaffected. The news coefficient is the slope of the winsorized spread on ν after demeaning both within cells defined by the debt choice and the income state, scaled by one standard deviation of ν. It is the model analogue of a panel regression of spreads on forecast revisions with nonparametric controls for debt and output. All statistics condition on market access.

identical income paths the rational economy's hazard is 1.5 percent. The mechanics are exactly Propositions 2 and 4 in sequence: the boom priced debt into the fragility corridor, and the first negative realization swings the belief tilt from extrapolated improvement to extrapolated decline, collapsing the price schedule and expanding the default set while debt is at its peak. Defaults follow booms in this model through fragile pricing created by the boom itself.

6.6 A short-bond fit diagnostic

A two-parameter exercise measures how the short-bond mechanism responds when default frequency and debt service return to their data values. At $\theta = 0.5$, selecting (β, ϕ) to match three defaults per century and debt service of 5.5 percent gives $\beta = 0.969$ and $\phi = 0.91$. The median spread becomes 0.1 percent. Maturity, spread volatility, and returns remain outside this two-target calculation and enter through the long-bond experiment.

At the selected pair, the within-cell news coefficient is -84 basis points per standard deviation and remains zero under rational pricing at the same parameters. The reversal-quarter hazards are 3.1 percent and 0.3 percent. The conditional news mechanism therefore persists when the short-bond default frequency and debt service return to their target values. Section 6.2 supplies the long-duration comparison.

Whether actual sovereign spreads behave this way is the next question.

7 Evidence from Spreads and Forecasts

Future bond returns organize the evidence. Diagnostic creditors pay too much after favorable revisions and earn low subsequent returns under the true probability law. The conditional spread slope then ties that return pattern to the model's history state. Asymmetry near the default

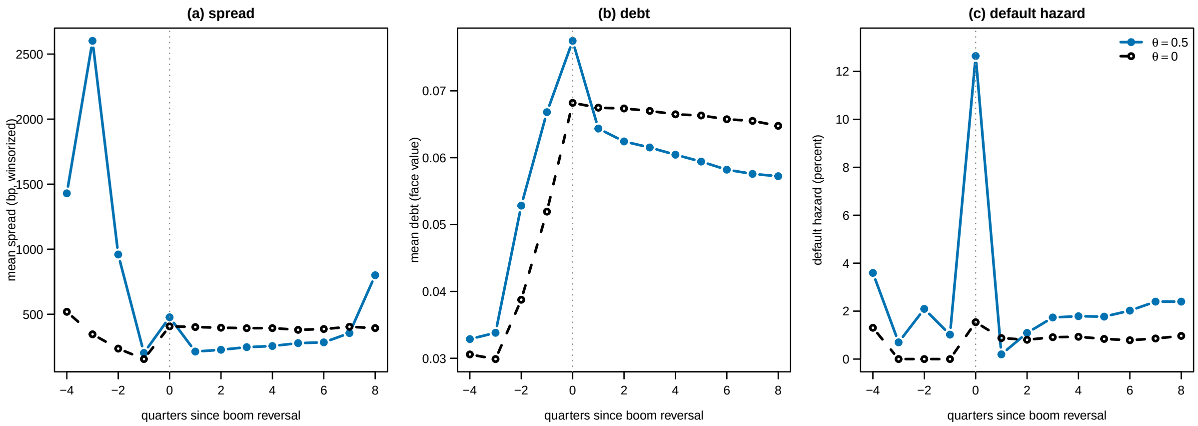

Figure 3: Boom-reversal episodes: at least three consecutive rises in log income followed by a fall on the common simulated income path (5,593 episodes). Quarter 0 is the reversal. Spread and debt paths condition on continuous market access through the window. The full risk set determines the default hazard.

boundary and cross-country variation in θ_i sharpen the mechanism. Table 8 in Appendix D records the data sources and coverage.

P1: favorable revisions predict low excess returns. Let $RX_{i,t+h}$ denote the cumulative EMBI total return in excess of a duration-matched United States Treasury return. The primary regression is

$$RX_{i,t+h} = \alpha_i + \tau_t + \beta_R \text{Rev}_{i,t} + \eta s_{i,t} + \Gamma' X_{i,t} + \varepsilon_{i,t+h}. \tag{14}$$

The diagnostic prediction is $\beta_R < 0$. The current spread absorbs the market's starting yield. The controls include duration, ratings, commodity terms of trade, global risk, and fiscal revisions. Local projections trace the return reversal from three to twenty-four months. The moving-block country bootstrap resamples both the forecast first stage and the overlapping return horizon.

P2: news moves spreads conditional on the model state. The estimating equation, at country i and month t,

$$s_{i,t} \;=\; \alpha_{c(i,t)} + \tau_t + \delta\, \text{Rev}_{i,t} + \Gamma' Z_{i,t} + e_{i,t}, \tag{15}$$

regresses the EMBI stripped spread on the consensus forecast revision of twelve-month-ahead real growth. The fixed effect $\alpha_{c(i,t)}$ indexes country-specific cells of debt and output relative to trend. Time effects absorb global conditions. The vector Z contains the forecast level, ratings, commodity terms of trade, and fiscal revisions. This flexible conditioning implements the model state. Inside the maintained rational default model, the conditional news coefficient equals zero. Broader rational models can load on reserves, maturity walls, political risk, and permanent growth news, so the return regression in (14) carries the general overreaction test. Equation (15) tests the sovereign mechanism that links overreaction to the default state.

P3: the price response rises near the default boundary. Within the model, positive

Table 5: Empirical predictions.

Test	Rational null	Model at $\theta = 0.5$	Estimation
P1: return predictability from news	0	negative after good news	EMBI total returns
P2: within-cell news coefficient	model-specific 0	-462 bp per 1σ	EMBI and forecasts
P3: boundary interaction	0	stronger near default	EMBI and forecasts
P4: response increasing in θ_i	flat relation	positive relation	cross-fitted country estimates
CG regression, individual forecasts	0	-0.30	published for US[a]

Notes: [a]Bordalo et al. (2020) report negative individual-level forecast-error-on-revision coefficients for most United States macroeconomic series. Section 5.2 specifies the emerging-market estimation. Data requirements: J.P. Morgan EMBI Global stripped spreads (Bloomberg or Datastream), Consensus Economics individual forecaster panel, IMF WEO vintages, CBOE VIX, sovereign ratings. Table 8 records access and replication files.

and negative news produce responses of -657 and -368 basis points per standard deviation. Interacting news with debt raises the response where repayment is marginal. The empirical specification adds Rev^+, Rev^-, and $\text{Rev} \times \text{Debt}$ to (15). The same interaction enters (14) with the opposite payoff implication.

P4: the measured belief parameter travels across data sets. The forecast block estimates θ_i from one cross-fitting fold. A disjoint fold estimates the return and spread responses. The model predicts stronger return reversal and a steeper conditional price response where θ_i is larger. The hierarchical second stage integrates over the first-stage distribution of θ_i rather than treating a noisy generated regressor as data.

Table 5 collects the designs. The first test establishes overpricing. The next two locate it in the sovereign default mechanism. The last links the market response to a belief parameter measured outside bond prices.

8 Fiscal Rules under Diagnostic Pricing

The instrument I study is the simplest one in the policy debate: a ceiling $\bar{b}$ on debt issuance, imposed permanently and known to all parties. For each ceiling on a grid I recompute the full equilibrium, price schedule included, and evaluate welfare as the value at zero debt and median income *under the true income process*, converting differences to basis points of permanent consumption. Because Section 2 taught that the answer turns on whose beliefs are distorted, I compute the complete two-by-two: government rational or diagnostic, market rational or diagnostic, with $\theta = 0.5$ wherever beliefs are diagnostic. A diagnostic government maximizes under its own tilted kernel (the Bellman equations are in Appendix E), and its welfare is then

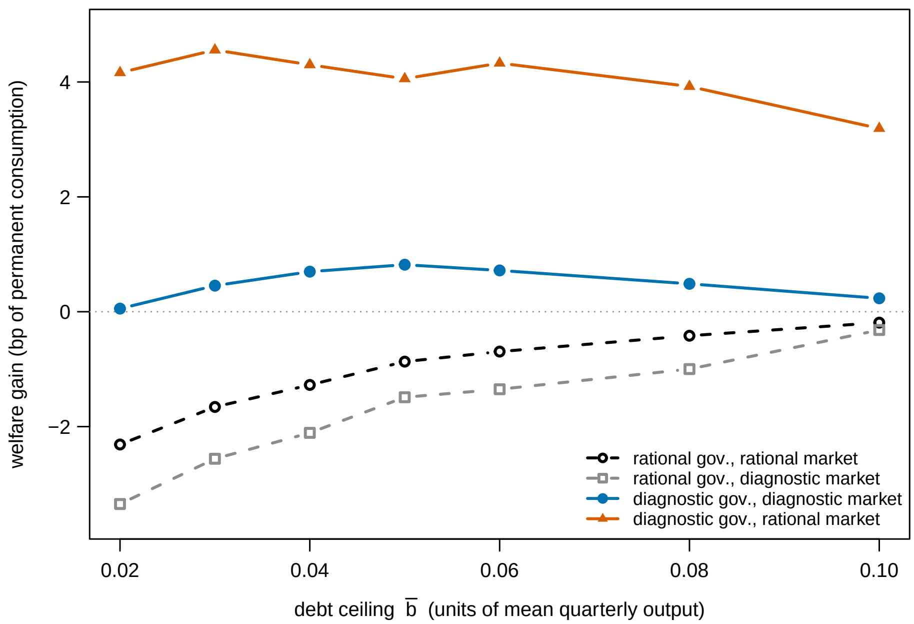

Figure 4: Welfare effect of a permanent debt ceiling $\bar{b}$, by configuration of beliefs, evaluated under the true income process at zero debt and median income. $\theta = 0.5$ where diagnostic.

Table 6: The whose-beliefs two-by-two: welfare gain of the best debt ceiling.

	Market rational	Market diagnostic
Government rational	-0.2 bp	-0.3 bp
Government diagnostic	+4.6 bp	+0.8 bp

Notes: basis points of permanent consumption, at the best ceiling on the grid $\{0.02, \ldots, 0.10\}$ per cell. Negative entries mean every ceiling on the grid lowers welfare. $\theta = 0.5$ in diagnostic cells.

evaluated under the truth. Everything in this section is a numerical property of computed equilibria at the baseline calibration, conditional on that welfare criterion. Figure 4 plots the welfare gain of each ceiling in each cell, and Table 6 reports the best ceiling per cell.

Sovereign welfare is the policy criterion. I evaluate induced government consumption under the true income law and report lender returns separately. A global criterion would place foreign marginal utility and international spillovers in the same units as sovereign consumption. The policy rankings below therefore concern sovereign welfare under the true probability measure.

The sign pattern is the result. When the government is rational, every ceiling hurts, and it hurts *more* when the market is diagnostic (-3.3 basis points at the tightest ceiling, against -2.3 in the fully rational cell): the rational government was selling overpriced debt to optimists, and the rule takes the trade away. When the government is diagnostic, ceilings help. Facing a diagnostic market the best ceiling yields +0.8 basis points and cuts the default frequency from 6.6 to 5.7 per century. Facing a rational market it yields +4.6. The gains are largest in the cell where the market is rational because there the government's extrapolation earns zero offsetting rents: rational lenders price the optimistic government's future behavior correctly, so its overborrowing (mean debt service 6.1 percent of output, against 4.4 in the rational benchmark) is pure loss, and the rule claws much of it back. Note also the diagnostic-government economies confirm the boom-overborrowing channel the two-period example leaves to the dynamic model: with persistent income, a government that extrapolates good news borrows against an expected future that disappoints, which is precisely the borrowing a good-news-contingent constraint would target.

The welfare magnitudes are single basis points of permanent consumption because the one-period economy has modest default costs and inexpensive consumption smoothing. The sign is the main result. Under one-period debt, a benevolent government, and the computed equilibria at $\theta = 0.5$, a debt ceiling lowers sovereign welfare when the government prices the future correctly and raises it when the government extrapolates. Fiscal rules act as commitment against the government's extrapolation. Forecast revisions reveal the relevant state in real time and support a rule that tightens after sustained favorable revisions.

9 Conclusion

Diagnostic expectations give sovereign debt an observable history state. The one-period model proves that prices rise with favorable news at fixed debt and income, true-measure returns fall, and default regions expand when the news reverses. The monthly survey model measures the same extrapolation in individual forecasts. The five-year bond economy preserves the exact rational history zero and quantifies the diagnostic effects with coupons, dilution, and capital gains. Rational intermediary capacity attenuates the news slope continuously.

Future excess returns provide the central empirical restriction. Favorable forecast revisions should predict low duration-adjusted sovereign bond returns, particularly near the default boundary and in countries with larger forecast-based estimates of θ. The within-state spread slope locates that overpricing in the sovereign default mechanism. Together, the forecast and return blocks distinguish diagnostic pricing from a calibration that matches spreads through default risk alone.

The policy result depends on the government's probability law. A rational government exploits creditor optimism and loses from a debt ceiling. A government that extrapolates uses the ceiling as a commitment device against its own borrowing plans. This division between market mispricing and government beliefs is the relevant distinction for fiscal rules.

A Proofs

A.1 Proof of Lemma 1

Write $m_x = \rho x$ and $m_h = \rho h$. Substituting the two Gaussian densities into (6) gives the unnormalized exponent

$$-\frac{(1+\theta)(x'-m_x)^2 - \theta(x'-m_h)^2}{2\sigma_\varepsilon^2}.$$

Define $m_\theta = (1+\theta)m_x - \theta m_h$. Expanding both squares and collecting terms yields

$$\begin{aligned}
&(1+\theta)(x'-m_x)^2 - \theta(x'-m_h)^2 \\
&\quad = x'^2 - 2m_\theta x' + (1+\theta)m_x^2 - \theta m_h^2 \\
&\quad = (x'-m_\theta)^2 + (1+\theta)m_x^2 - \theta m_h^2 - m_\theta^2.
\end{aligned}$$

The last three terms do not depend on x', so the normalizing constant removes them. The normalized density is therefore

$$\hat{p}_\theta(x' \mid x, h) = \frac{1}{\sqrt{2\pi\sigma_\varepsilon^2}} \exp\left[-\frac{(x'-m_\theta)^2}{2\sigma_\varepsilon^2}\right].$$

Finally,

$$m_\theta = (1+\theta)\rho x - \theta\rho h = \rho x + \theta\rho(x-h),$$

which proves the stated mean and variance. ■

A.2 Proof of Proposition 1

Let $\mathcal{S}$ be the finite set of market-access and exclusion states. For each market-access state $s \in \mathcal{S}$, let $\mathcal{A}(s)$ contain default and every debt-grid choice. Utility below the consumption floor is extended by the tangent at the floor. This extension is finite, continuous, strictly increasing, and agrees with CRRA utility above the floor. The product of the mixed-action simplexes,

$$\Sigma = \prod_{s\in\mathcal{S}} \Delta(\mathcal{A}(s)),$$

is nonempty, compact, and convex. The price set

$$\mathcal{Q} = [0, 1/R]^{|\mathcal{B}||\mathcal{X}|^2}$$

has the same three properties.

Fix a price vector $q \in \mathcal{Q}$. The government's problem is a finite discounted Markov decision problem. Current utility is bounded on the finite action-state set, the transition matrix is stochastic, and $0 < \beta < 1$. Its Bellman operator is therefore a contraction in the sup norm with modulus β, so it has a unique value vector $V(q)$. Because every action value is a finite sum of continuous functions of q and $V(q)$, the value vector is continuous in q. At each state, the set of optimal mixed actions is the simplex supported on the maximizing pure actions. The product correspondence $\Gamma(q) \subseteq \Sigma$ is consequently nonempty, compact, convex valued, and upper hemicontinuous.

For a mixed stationary policy $\sigma \in \Sigma$, let $P(\sigma) \in \mathcal{Q}$ be the zero-profit price vector. Each component is $1/R$ times a finite sum of diagnostic transition probabilities multiplied by the next-state mixed repayment probability. The map P is linear in σ and hence continuous.

Define the correspondence on $\mathcal{Q} \times \Sigma$ by

$$\mathcal{F}(q, \sigma) = \{P(\sigma)\} \times \Gamma(q).$$

Its domain is nonempty, compact, and convex. Its values are nonempty, compact, and convex. Continuity of P and upper hemicontinuity of Γ imply that $\mathcal{F}$ is upper hemicontinuous. Kakutani's fixed-point theorem therefore gives $(q^*, \sigma^*) \in \mathcal{F}(q^*, \sigma^*)$. The policy σ^* is optimal at q^* and q^* satisfies zero profit under σ^*, which is a stationary recursive equilibrium in mixed policies. If a computed deterministic policy has zero Bellman and pricing residuals, its degenerate mixed representation satisfies the same two conditions in the extended economy. If every repayment choice gives consumption above the floor, the extension is never used at an optimal action. The deterministic fixed point then solves the original positive-consumption problem. ■

A.3 Proof of Lemma 2

By Lemma 1, $F_\theta(\cdot \mid x, h_i)$ is the Gaussian cdf with variance σ_ε^2 and mean $\mu_i = \rho x + \theta \rho \nu_i$. For $\nu_2 > \nu_1$ and $\theta > 0$, $\mu_2 > \mu_1$, and a Gaussian family with common variance is strictly ordered by first-order stochastic dominance in its mean: $F(z - \mu)$ is decreasing in μ for every z. ■

A.4 Proof of Lemma 3

Fix (x, h) and debt levels $b_2 > b_1$ in $\mathcal{B}$. If repayment is feasible at b_2, compactness of $\mathcal{B}$ and continuity on the feasible set give a maximizing issuance choice b_2'. Consumption from that same choice at b_1 exceeds consumption at b_2 by $b_2 - b_1 > 0$. The continuation value is unchanged because it depends on b_2' and (x, h) but not on inherited debt. Strict monotonicity of u therefore gives

$$\begin{aligned} V^R(b_1, x, h) &\geq u(y + q(b_2', x, h) b_2' - b_1) + \beta \mathbb{E}[V(b_2', x', x) \mid x] \\ &> u(y + q(b_2', x, h) b_2' - b_2) + \beta \mathbb{E}[V(b_2', x', x) \mid x] \\ &= V^R(b_2, x, h). \end{aligned}$$

If repayment is infeasible at b_2, default is forced and the upper-set conclusion already holds at that debt. The default value in (9) is independent of inherited debt. Hence $d = \mathbf{1}\{V^D > V^R\}$ is nondecreasing in b, and the default set is an upper interval. ■

A.5 Proof of Lemma 4

With i.i.d. income, the price schedule and both continuation values are independent of current income at fixed inherited debt and belief state. For each feasible issuance choice b', write the repayment advantage over default as

$$G_{b'}(y) = u\Big(c^R(y, b')\Big) + \beta C(b') - u((1-\psi)y) - \beta C^D.$$

Current repayment consumption increases one for one with income. Therefore

$$G'_{b'}(y) = u'\Big(c^R(y, b')\Big) - (1-\psi)u'((1-\psi)y)\,.$$

Concavity of utility and $c^R(y, b') \le \bar{c}(y)$ imply

$$u'\Big(c^R(y, b')\Big) \ge u'(\bar{c}(y)) \ge (1-\psi)u'((1-\psi)y)\,.$$

Thus every $G_{b'}$ is nondecreasing. Since

$$V^R(b, y) - V^D(y) = \max_{b'} G_{b'}(y),$$

the repayment advantage is nondecreasing as well. Under CRRA utility, substituting $u'(c) = c^{-\gamma}$ gives the stated consumption bound. Under linear utility, the derivative is $1 - (1-\psi) = \psi > 0$. ■

A.6 Proof of Proposition 2 and Corollary 1

Fix (b', x) and write $g(x') = 1 - d(b', x', x)$, the repayment indicator next period. Under Assumption 1, g is nondecreasing. From (10), $R\,q = \int g\,dF_\theta(\cdot \mid x, h)$. If $\nu_2 > \nu_1$, Lemma 2 gives first-order dominance, and the expectation of a nondecreasing function is nondecreasing under first-order dominance. Hence q is nondecreasing in ν.

For strictness, suppose $\theta > 0$ and default risk is interior at (b', x): $0 < \int g\,dF_\theta < 1$. Because income is Markov on a totally ordered set and g is a nondecreasing indicator, interiority means g switches from 0 to 1 at some finite threshold $\underline{x}'(b', x)$ in the interior of the support. Writing $Z \sim N(0, \sigma_\varepsilon^2)$,

$$R\,q(\nu) = \mathbb{E}\big[\,g(\mu(\nu) + Z)\,\big] = \mathbb{P}\{Z \ge \underline{x}' - \mu(\nu)\},$$

with $\mu(\nu) = \rho x + \theta\rho\nu$ strictly increasing in ν. Since the Gaussian distribution assigns positive density everywhere, the probability is strictly increasing in μ. This proves the proposition.

For the corollary: at $\theta = 0$ the kernel $\hat{p}_0(\cdot \mid x, h) = p(\cdot \mid x)$ is independent of h, so q is a function of (b', x) alone and any statistic of q computed within a (b', x) cell has zero variation in ν. Conversely, if $\theta > 0$ and default risk is interior at some (b', x) visited with positive probability, the strict monotonicity just established makes the within-cell slope strictly negative for spreads (spreads are strictly decreasing transformations of q). ■

A.7 Proof of Proposition 3

A lender pays $q(b', x, h) > 0$ and receives $g(x') = 1 - d(b', x', x)$. At fixed (b', x), define $A = \mathbb{E}_p[g \mid x]$ and $B(\nu) = \mathbb{E}_{\hat{p}_\theta}[g \mid x, h]$. The true-measure expected excess gross return is

$$R^e(\nu) = R\left(\frac{A}{B(\nu)} - 1\right).$$

The quantity A does not depend on ν. Proposition 2 implies that $B(\nu)$ is positive and nondecreasing on the positive-price domain. Thus, if $\nu_2 > \nu_1$, then $A/B(\nu_2) \le A/B(\nu_1)$, so R^e is nonincreasing. At $\theta = 0$, the diagnostic and true kernels coincide, which gives $A = B$ and $R^e = 0$. If $\theta > 0$ and $\nu > 0$, the diagnostic kernel first-order stochastically dominates the true kernel. Since g is nondecreasing, $B(\nu) \ge A$ and $R^e \le 0$. For $\nu < 0$, the true kernel dominates the diagnostic kernel, so $B(\nu) \le A$ and $R^e \ge 0$. The inequalities are strict when the repayment indicator changes on a set with positive diagnostic probability. ■

A.8 Proof of Proposition 4

Take the equilibrium price schedule q and continuation value as given, and compare two histories $\nu_2 > \nu_1$ at the same (b, x). For every candidate issuance $b' \ge 0$, consumption $y + q(b', x, h)b' - b$ is nondecreasing in ν by Proposition 2, and the continuation $\mathbb{E}[V(b', x', x) \mid x]$ is independent of h because the government's expectation is rational and tomorrow's reference is today's x. Hence each candidate's objective in (8) is nondecreasing in ν, and V^R, a pointwise supremum of nondecreasing functions of ν, is nondecreasing in ν. V^D in (9) involves neither b nor h. Therefore $d = \mathbf{1}\{V^D > V^R\}$ is nonincreasing in ν, which is the set inclusion claimed. ■

A.9 Proof of Proposition 5

Under the fixed-target convention, the diagnostic forecast of x_{t+1} made at t is $F_t x_{t+1} = \mathbb{E}_t x_{t+1} + \theta(\mathbb{E}_t x_{t+1} - \mathbb{E}_{t-1} x_{t+1})$ with $\mathbb{E}_t x_{t+1} = \rho x_t$ and $\mathbb{E}_{t-1} x_{t+1} = \rho^2 x_{t-1}$, so

$$F_t x_{t+1} = \rho x_t + \theta\rho\varepsilon_t, \qquad F_{t-1} x_{t+1} = \rho^2 x_{t-1} + \theta\rho^2\varepsilon_{t-1}.$$

Hence the revision and the error are

$$\mathrm{Rev}_t = F_t x_{t+1} - F_{t-1} x_{t+1} = \rho(1+\theta)\varepsilon_t - \theta\rho^2\varepsilon_{t-1}, \qquad \mathrm{FE}_{t+1} = x_{t+1} - F_t x_{t+1} = \varepsilon_{t+1} - \theta\rho\,\varepsilon_t.$$

The innovations are i.i.d., so

$$\mathrm{Cov}(\mathrm{FE}_{t+1}, \mathrm{Rev}_t) = -\theta(1+\theta)\rho^2\sigma_\varepsilon^2, \qquad \mathrm{Var}(\mathrm{Rev}_t) = \rho^2\sigma_\varepsilon^2\big[(1+\theta)^2 + \theta^2\rho^2\big],$$

and the ratio is (12). The same algebra at horizon k for a fixed target x_{t+k} gives $\mathrm{Rev} = \rho^k(1+\theta)\varepsilon_t - \theta\rho^{k+1}\varepsilon_{t-1}$ and $\mathrm{FE} = \sum_{j=1}^k \rho^{k-j}\varepsilon_{t+j} - \theta\rho^k\varepsilon_t$. The coefficient is horizon invariant for that fixed-target experiment. Annual fixed-event forecasts combine targets and calendar weights.

They require the simulation mapping in Section 5.2. ■

B The Two-Period Example: Details

Consumption at date 0 is $c_0 = y_0 + qb$ with $q = \hat{\pi}/R$ in the risky region $\psi y_L < b \le \psi y_H$. The objective is $\ln c_0 + \beta[\pi \ln(y_H - b) + (1-\pi)\ln((1-\psi)y_L)]$, and only the first two terms involve b. The first-order condition is

$$\frac{\hat{\pi}/R}{y_0 + (\hat{\pi}/R)b} = \frac{\beta\pi}{y_H - b},$$

and solving the linear equation gives (4). Differentiating (4) in $\hat{\pi}$ gives $\partial b^*/\partial\hat{\pi} = \beta\pi R y_0/[\hat{\pi}^2(1+\beta\pi)] > 0$. The second-order condition holds by concavity in b.

The safe–risky margin. The best safe plan solves $\max_{b \le \psi y_L} \ln(y_0 + b/R) + \beta[\pi \ln(y_H - b) + (1-\pi)\ln(y_L - b)]$ and its value W^S is independent of $\hat{\pi}$. The best risky plan's value $W^R(\hat{\pi})$ is strictly increasing in $\hat{\pi}$ by the envelope theorem, since $\partial W^R/\partial\hat{\pi} = u'(c_0)\, b^*/R > 0$. If parameters are such that $W^R(\pi) < W^S < \lim_{\hat{\pi}\to 1} W^R(\hat{\pi})$, there is a threshold $\hat{\pi}^\dagger$ at which the sovereign switches from safe to risky borrowing, and optimism beyond the threshold raises the true default probability from 0 to $1-\pi$ discontinuously.

A diagnostic government in the example. If the government also prices the future with $\hat{\pi}$, the first-order condition becomes $(\hat{\pi}/R)/c_0 = \beta\hat{\pi}\, u'(y_H - b)$, in which $\hat{\pi}$ cancels from the price side and the belief side:

$$\hat{b} = \frac{y_H - \beta R y_0}{1 + \beta\hat{\pi}},$$

which is *decreasing* in $\hat{\pi}$. Probability optimism makes repayment subjectively more likely and so subjectively more expensive, and with log utility this cost effect outweighs the price effect. The example is therefore the probability-optimism case. In the infinite-horizon model, optimism is about the *level* of future income, and the consumption-smoothing motive of an optimist is to borrow against the good future it expects. Quantitatively that channel dominates, and mean debt in the diagnostic-government cells exceeds the rational benchmark (Section 8). The contrast disciplines the interpretation of two-period intuition in this class of models.

C Computation

Discretization. Log income uses a Tauchen (1986) chain with $n_x = 21$ states spanning ± 3 unconditional standard deviations. The diagnostic kernel is discretized by the same binning rule applied to the $N(\rho x_i + \theta\rho(x_i - x_k),\, \sigma_\varepsilon^2)$ density for every pair (i,k), which is exact up to the binning because Lemma 1 keeps the tilted density Gaussian with unchanged variance. Debt lives on an evenly spaced grid of $n_b = 200$ points on $[0, 0.35]$ in units of mean quarterly output.

Algorithm. Given a price array q, one backward-induction sweep updates V^D, V^R (by full maximization over the debt grid), V, and d. The price is then updated from (10) and damped, $q \leftarrow \frac{1}{2}q + \frac{1}{2}q^{\text{new}}$. The joint iteration runs to sup-norm tolerances of 5×10^{-8} on both V and q, reached in 351 iterations at every θ, and a full solve takes roughly twenty seconds in multithreaded Julia on a laptop. At $\theta = 0$ the computed equilibrium reproduces the Arellano (2008) benchmark by construction.

Simulation. 500,000 quarters after a 1,000-quarter burn-in, with the income path drawn once and reused across every configuration (common random numbers), so that cross-θ comparisons and the event study condition on identical fundamentals. Statistics condition on market access. Spreads are annualized from quarterly prices and winsorized at the 1st and 99th percentiles for means and standard deviations.

Survey mapping. The monthly validation uses 240000 simulated months. At each date the code constructs current-year and next-year annual-average growth forecasts, converts them to a rolling twelve-month forecast with the empirical calendar weights, and estimates the same forecast-error regression used on the data. The code evaluates the map on a grid of θ values and inverts it by monotone bisection. This routine recovers inputs $\theta \in \{0, 0.25, 0.5, 1\}$ within 0.02. At the baseline, the recovery error is 0.0.

Verification of Assumption 1 and grid robustness. In the computed equilibria at $\theta \in \{0, 0.5\}$ income-monotonicity and the debt-monotonicity that Lemma 3 guarantees hold across all grid combinations. At $\theta = 1$ a small fraction of grid columns shows a violation at extreme states with near-degenerate pricing: 0.02 percent at the baseline grids and 0.1 percent at doubled grids. Doubling both grids ($n_x = 31$, $n_b = 400$) gives zero violations at $\theta \in \{0, 0.5\}$. Replacing the asymmetric default cost with a proportional cost $y^D = (1 - \psi)y$ for $\psi \in \{0.02, 0.05\}$ gives zero violations. All thirty parameter vectors visited by the recalibration search, and the recalibrated final pair, satisfy both monotonicities at every grid point. The headline moments are also stable under grid doubling: at $\theta = 0.5$ the within-cell news coefficient moves from -462 to -400 basis points per standard deviation, the default frequency from 11.6 to 11.4 per century, and the reversal-quarter hazard from 12.6 to 12 percent (1.5 to 1.1 for the rational economy), while the rational news coefficient remains zero (0). The winsorized spread standard deviation is visibly grid-sensitive (57.8 against 35.3 percent), as it inherits the extreme spread tail. The volatility ratio to the rational economy remains far above one at both densities, and the paper's conclusions use the news coefficient, default frequency, and reversal hazard.

Short-bond diagnostic grid. The grid behind Section 6.6 is

$$\beta \in \{0.945, 0.953, 0.961, 0.969, 0.977\}, \qquad \phi \in \{0.880, 0.895, 0.910, 0.925, 0.945, 0.969\}.$$

Table 7: Parameter discipline and sensitivity.

Parameter	Units, frequency	Category	Source or target (sample)	Range examined	Effect on headline results
$\gamma = 2$, eq. (8)	dimensionless	externally fixed	standard value in the quantitative default literature	fixed	scales welfare units, signs fixed
$\beta = 0.953$, eq. (8)	quarterly	internally calibrated	default moments at $\theta = 0$ (Arellano, 2008), re-selected at $\theta = 0.5$: 0.969	0.945–0.977	default frequency falls steeply in β (14.2 to 1.0 per century across the grid)
$r^* = 0.017$, eq. (10)	quarterly	externally fixed	U.S. five-year yield (Arellano, 2008)	fixed	shifts spread levels one for one
$\rho = 0.945$, eq. (5)	quarterly	directly estimated	detrended Argentine real GDP (Arellano, 2008)	mapping check	$\beta_{CG}(\theta)$ mapping nearly flat in ρ, tilt size scales with $\theta\rho$
$\sigma_\varepsilon = 0.025$, eq. (5)	quarterly	directly estimated	detrended Argentine real GDP (Arellano, 2008)	fixed	scales news size, signs fixed
$\lambda = 0.282$, eq. (9)	quarterly	externally fixed	market re-access delays (Arellano, 2008)	fixed	shortens exclusion, lowers effective default cost
$\phi = 0.969$, eq. (9)	fraction of $\mathbb{E}[y]$	internally calibrated	spread moments at $\theta = 0$ (Arellano, 2008), re-selected at $\theta = 0.5$: 0.91	0.880–0.969	governs default frequency and debt capacity jointly with β
$\theta = 0.5$, eq. (6)	dimensionless	baseline in published range	individual CG regressions (Bordalo et al., 2020), EM estimate in Section 5.2	$\{0, 0.25, 0.5, 1\}$	mechanism vanishes exactly at $\theta = 0$, market access collapses near $\theta = 1$ at fixed (β, ϕ)

Notes: categories follow the four-way scheme of Section 6. Validation uses moments separate from calibration: the within-cell news coefficient, the return-predictability pattern, and the boom-reversal hazard profile all stay outside the calibration and recalibration loss. The region where the mechanism disappears is stated explicitly: all history dependence vanishes at $\theta = 0$, continuously, and the environment stops resembling emerging-market data near $\theta = 1$ at fixed parameters outside beliefs.

The solver computes each pair at $\theta = 0.5$ and simulates 200,000 quarters. This short-bond exercise targets default frequency and debt service. The long-duration model retains the published joint calibration described in Section 6.2.

Welfare. Consumption equivalents between configurations use $\Delta = (W_1/W_0)^{1/(1-\gamma)} - 1$ evaluated at zero debt and median income under the true measure. Policy evaluation computes W for diagnostic-government cells as in Appendix E.

D Data Appendix

Spreads. J.P. Morgan EMBI Global stripped spreads, by country, monthly averages of daily quotes, accessed through Bloomberg or Datastream. The stripped spread nets out collateralized cash flows and is the standard market-risk measure in this literature (Uribe and Yue, 2006; Longstaff et al., 2011). Spreads are winsorized at the 1st and 99th percentiles within country. Default and restructuring windows are coded using the Cruces and Trebesch (2013) episode list and excluded from pricing regressions, mirroring the model's conditioning on market access.

Forecasts. Consensus Economics provides monthly, individual-forecaster surveys of real GDP growth for the current and next calendar year in the major emerging markets from the early-to-mid 1990s. The code converts fixed-event forecasts to a fixed twelve-month horizon by weighting the two events by months remaining in the year: $F_t^{12m} = \frac{m}{12} F_t^y + (1 - \frac{m}{12}) F_t^{y+1}$, with m the months left in year y. Month-over-month changes in F^{12m} give revisions. Within-event-year changes give the fixed-target revisions for the Coibion–Gorodnichenko regressions. The IMF World Economic Outlook vintage database supplies the *first-release* growth outturn, so forecast errors use the information set available before later data revisions.

Fundamentals and controls. General government debt to GDP comes from the WEO, and output relative to trend from a country-specific quadratic trend in log real GDP (robustness: Hamilton filter). The CBOE VIX and the ten-year United States Treasury yield serve as global controls, absorbed by time effects in the baseline. Standard & Poor's long-term foreign-currency ratings are mapped to a 21-notch scale. One timing mismatch deserves notice: WEO debt is annual, and interpolating it to monthly frequency puts measurement error inside the conditioning cells. The design therefore uses lagged debt for cell assignment to avoid look-ahead, and quarterly BIS debt data cover a subsample for robustness.

Sample. The intersection of EMBI Global membership and Consensus coverage with at least ten years of joint data, roughly 25 countries including Argentina, Brazil, Bulgaria, Chile, China, Colombia, Ecuador, Hungary, Indonesia, Malaysia, Mexico, Nigeria, Panama, Peru, the Philippines, Poland, Russia, South Africa, Turkey, Ukraine, and Venezuela, monthly, 1994–2024. Cell fixed effects for (15) use country-specific terciles of debt-to-GDP crossed with terciles of output relative to trend, with finer bins as a robustness dimension. Standard errors are clustered two-way by country and month, with Driscoll–Kraay errors as robustness given overlapping forecast horizons.

E Extensions

E.1 The diagnostic government

When the government itself holds diagnostic beliefs, its expectations in (8)–(9) are taken under $\hat{p}_{\theta_g}(\cdot \mid x, h)$, and the default value acquires the belief state: the Bellman system becomes

$$V^R(b, x, h) = \max_{b'} \Big\{ u(y + q(b', x, h)b' - b) + \beta \int V(b', x', x)\, \hat{p}_{\theta_g}(x' \mid x, h)\, dx' \Big\}, \tag{E.1}$$

$$V^D(x, h) = u(y^D(x)) + \beta \int \left[\lambda V(0, x', x) + (1 - \lambda) V^D(x', x)\right] \hat{p}_{\theta_g}(x' \mid x, h)\, dx', \tag{E.2}$$

with lenders pricing under $\hat{p}_{\theta_\ell}$ as before. The formulation treats the government as sophisticated about its own future beliefs in the stationary sense: it correctly anticipates that its future self will evaluate continuation problems with the tilt implied by the future state (x', x). Because the tilted

Table 8: Data and implementation.

Dataset	Coverage	Frequency	Access	Observed or proxied	Replication
EMBI Global stripped spreads (J.P. Morgan)	∼25 countries, 1994–2024	daily, used monthly	Bloomberg or Datastream license	observed	code shareable, raw data licensed
Consensus Economics, individual forecasters	major emerging markets, mid-1990s–2024	monthly	commercial license	observed	code shareable, raw data licensed
IMF WEO vintage database	all sample countries	semiannual vintages	public	observed	shareable
WEO debt to GDP	all sample countries	annual	public	interpolated to monthly (proxy)	shareable
S&P foreign-currency ratings	all sample countries	event dates	licensed	mapped to 21-notch scale	derived series shareable
CBOE VIX, U.S. Treasury yields	global	daily	public	observed	shareable
Cruces and Trebesch (2013) restructurings	episodes, 1970–2013	episode	public	observed	shareable

Notes: the two licensed panels supply spreads and individual forecasts. The replication archive contains all code, all public data, and exact extraction instructions for the licensed series.

kernel is Markov on (x, h), the fixed point is well defined. The alternative, anticipated-utility formulation in which the government treats today's tilted forecast as permanent gives similar quantitative answers in related settings and is a robustness item. *True* welfare of the induced policies (d, b') solves the linear system

$$W(b,x,h) = \begin{cases} W^D(x), & d(b,x,h) = 1, \\ u\big(c(b,x,h)\big) + \beta \int W\big(b'(b,x,h), x', x\big)\, p(x' \mid x)\, dx', & d = 0, \end{cases}$$

with W^D defined analogously under p, and all welfare statements in Section 8 use W.

E.2 Long-duration bonds

The quantitative model uses the random-maturity bond of Chatterjee and Eyigungor (2012). A fraction δ of each unit matures each quarter. The remaining fraction pays coupon z and survives. If b is debt at the start of the quarter and b' is debt after issuance, repayment consumption is

$$c = y + m - \big[\delta + (1-\delta)z\big]b + q(b', x, h)\big[b' - (1-\delta)b\big]. \tag{E.3}$$

The sovereign's repayment value is

$$V^R(b,x,h,m) = \max_{b' \in \mathcal{B}} \left\{ u(c) + \beta Z(b', x) \right\},$$

where

$$Z(b', x) = \sum_{x'} P(x' \mid x) \int V(b', x', x, m')\, dG(m').$$

The temporary shock has a truncated normal distribution. It is independent over time and has standard deviation 0.003, the value in Chatterjee and Eyigungor (2012). Default in the current quarter resets it to the lower support point. During exclusion it is again drawn from G. The code therefore keeps separate the integrated exclusion value and the value used in the current default comparison. Numerical stability comes from continuation in θ and damping of the fixed-point map. No economic parameter is altered to obtain convergence.

Competitive lenders satisfy

$$q(b', x, h) = \frac{1}{R}\widehat{\mathbb{E}}_\theta\Big[(1 - d(b', x', x, m'))\,\{\delta + (1-\delta)\,[z + q(b''(b', x', x, m'), x', x)]\}\;\Big|\; x, h\Big].$$

Thus the risk-free price is $[\delta + (1-\delta)z]/(\delta + r^*)$. Setting $\delta = 1$ removes the resale term and reproduces the one-period price and default arrays point for point in the numerical test.

Proposition 6 (The rational history zero with long debt)**.** *Set $\theta = 0$ and restrict attention to minimal-state Markov equilibria. The long-bond Bellman and pricing operators preserve the subspace in which values, policies, and prices are independent of h. Any fixed point reached in that subspace has $q(b', x, h) = q(b', x)$, so the within-cell news coefficient equals zero for every maturity and coupon. If the unrestricted fixed point is unique, every stationary Markov equilibrium has this property.*

Proof. At $\theta = 0$, the lender kernel depends on x and not on h. Suppose the current price and continuation objects are independent of h. The sovereign's budget, current utility, and rational continuation value then depend on (b, x, m) alone. Its optimal issuance and default rules inherit the same state space. Substituting these policies into the lender equation makes the payoff and its expectation independent of h. The joint operator therefore maps the history-invariant subspace into itself. Iteration from a history-invariant initial condition remains in that closed subspace, and every limit fixed point has the stated price schedule. Uniqueness extends the result to the unrestricted state representation. ■

For $\theta > 0$, the resale price and next-period issuance rule both respond to history. The short-bond FOSD proof no longer signs every term in the long-bond payoff. Table 3 therefore reports long-bond price monotonicity and return signs as computed equilibrium properties.

The continuous shock supplies the computational regularization used by Chatterjee and Eyigungor (2012). For every (b, x, h), the algorithm recovers all thresholds at which the optimal debt choice changes. It integrates bond payoffs exactly over the associated probability masses and integrates utility by twenty-node Gaussian quadrature within each fixed-policy interval. Adaptive quadrature provides an independent accuracy check. The fixed-point objects are Z, the integrated exclusion value, and q. Values receive full Bellman updates while prices use damping and safeguarded Anderson acceleration. Continuation in θ begins on a 12-node debt grid. Linear interpolation prolongs the target solution to 50 debt nodes, where the original operator is resolved. The largest continuation, exclusion, or pricing residual is 5.0e-8. Risk-free and zero price schedules lead to the same low-resolution equilibrium. The one-period limit reproduces the separate short-bond code. Near ties can generate deterministic policy cycles on intermediate grids. The mixed-policy existence result covers these ties, and the reported 50-node deterministic equilibrium satisfies the original fixed-point equations.

The economic parameter block comes from Chatterjee and Eyigungor (2012). It sets $\delta = 0.05$, $z = 0.03$, $r^* = 0.01$, the re-entry probability to 0.0385, and $\beta = 0.95460$. The persistent income process has $\rho = 0.948503$ and innovation standard deviation 0.027092. The default loss is $\max\{0, d_0 y + d_1 y^2\}$ with $d_0 = -0.18845$ and $d_1 = 0.24559$. That paper selected these values jointly to match mean debt, the mean spread, and spread volatility. Holding the block fixed gives the diagnostic comparison its controlled interpretation.

E.3 Capital-constrained rational intermediaries

Let $v_D(\nu)$ denote a diagnostic investor's discounted value of the long-bond payoff and let v_R denote the rational value. Diagnostic funds and rational intermediaries choose holdings n_D and n_R. A quadratic balance-sheet cost $\kappa_j n_j^2/2$ captures each sector's finite risk-bearing capacity. At an interior allocation,

$$n_D = \frac{v_D(\nu) - q}{\kappa_D}, \qquad n_R = \frac{v_R - q}{\kappa_R}.$$

Market clearing requires $n_D + n_R = B$. Solving for the common price gives

$$q(\nu) = \frac{\kappa_R v_D(\nu) + \kappa_D v_R - \kappa_D \kappa_R B}{\kappa_D + \kappa_R}.$$

Since the rational valuation is history invariant,

$$\frac{\partial q}{\partial \nu} = \omega_D \frac{\partial v_D}{\partial \nu}, \qquad \omega_D = \frac{\kappa_R}{\kappa_D + \kappa_R} \in (0, 1).$$

Greater rational risk-bearing capacity lowers κ_R and drives ω_D continuously toward zero. A tightly constrained rational sector has a small price impact. An unconstrained rational sector eliminates the diagnostic news slope. Nonnegativity constraints create the corresponding piecewise-linear corners without changing the interior attenuation result.

E.4 Reference-belief conventions

The model's reference density is $p(\cdot \mid h)$, the kernel evaluated at yesterday's state, which keeps the belief state two-dimensional and Lemma 1 exact. Bordalo et al. (2020) define the reference as the lagged forecast of the same target, giving the tilted mean $\rho x + \theta\rho(x - \rho h) = \rho x + \theta\rho\varepsilon_t$. The two means differ by $\theta\rho(1-\rho)h$, about five percent of the tilt at the calibrated persistence. Every result in Section 4 goes through under the alternative convention with ν redefined as the innovation ε. The FOSD ordering in Lemma 2 needs only a mean tilt increasing in news. The fixed-target horizon result in Proposition 5 does not by itself cover annual fixed-event forecasts.

Section 5.2 supplies that mapping by simulation.

E.5 Additional quantitative results

At $\theta = 1$ the one-period economy defaults 29.3 times per century and holds market access 73.9 percent of the time. Median spreads fall to 0.5 percent while the winsorized standard deviation rises sharply. This fixed-parameter stress test shows how sovereign forecast measurement disciplines the transfer of credit-cycle estimates across asset classes. Figure 4 reports the full ceiling grids behind Table 6. The optimal ceilings are $\bar{b} = 0.05$ and $\bar{b} = 0.03$ in the diagnostic-government cells. Replication code records wall time, iteration counts, residuals, and random seeds for every configuration.

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